



**CARIBBEAN EXAMINATIONS COUNCIL**

**CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®**

**PHYSICS**

**UNIT 2 – Paper 02**

*2 hours 30 minutes*

**READ THE FOLLOWING INSTRUCTIONS CAREFULLY.**

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. All working MUST be shown.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of figures in answers will be penalized.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
7. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.**

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# LIST OF PHYSICAL CONSTANTS

|                              |                             |   |                                         |
|------------------------------|-----------------------------|---|-----------------------------------------|
| Speed of light in free space | $c$                         | = | $3.00 \times 10^8 \text{ m s}^{-1}$     |
| Permeability of free space   | $\mu_0$                     | = | $4\pi \times 10^{-7} \text{ H m}^{-1}$  |
| Permittivity of free space   | $\epsilon_0$                | = | $8.85 \times 10^{-12} \text{ F m}^{-1}$ |
|                              | $\frac{1}{4\pi \epsilon_0}$ | = | $9.0 \times 10^9 \text{ m F}^{-1}$      |
| Elementary charge            | $e$                         | = | $1.60 \times 10^{-19} \text{ C}$        |
| Planck's constant            | $h$                         | = | $6.63 \times 10^{-34} \text{ J s}$      |
| Unified atomic mass constant | $u$                         | = | $1.66 \times 10^{-27} \text{ kg}$       |
| Energy equivalence           | $1 u$                       | = | $931 \text{ MeV}/c^2$                   |
| Rest mass of electron        | $m_e$                       | = | $9.11 \times 10^{-31} \text{ kg}$       |
| Rest mass of proton          | $m_p$                       | = | $1.67 \times 10^{-27} \text{ kg}$       |
| Acceleration due to gravity  | $g$                         | = | $9.81 \text{ m s}^{-2}$                 |
| 1 Atmosphere                 | atm                         | = | $1.00 \times 10^5 \text{ N m}^{-2}$     |
| Avogadro's constant          | $N_A$                       | = | $6.02 \times 10^{23} \text{ per mole}$  |

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SECTION A

Answer ALL questions.

Write your answers in the spaces provided.

1. (a) Figure 1 shows a coil of wire made from an unknown metal.

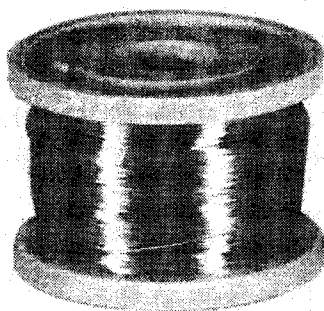


Figure 1. Resistance coil

Distinguish between the 'electrical resistance' and the 'electrical resistivity' of the coil of wire.

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[2 marks]

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- (b) Identify the circuit shown in Figure 2.

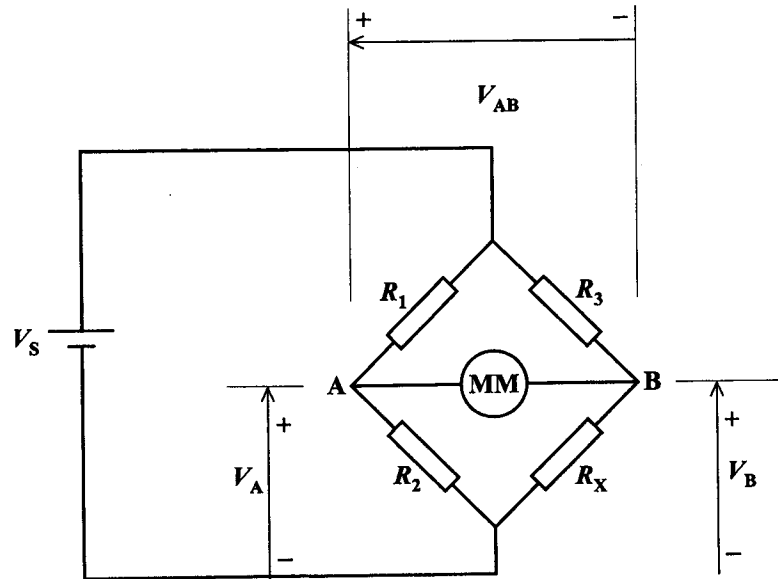


Figure 2. A circuit

[1 mark]

- (c) By treating the circuit in Figure 2 as two potential dividers in parallel, show that the voltage,  $V_{AB}$ , indicated by the multimeter (MM) will be given by:

$$V_{AB} = \left( \frac{R_2}{R_1 + R_2} - \frac{R_x}{R_x + R_3} \right) V_s$$

[4 marks]

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- (d) The circuit in Figure 2 is set up with the following component values:

$$R_1 = R_3 = 4.0 \text{ k}\Omega$$

$$R_2 = 1.0 \text{ k}\Omega$$

$$V_s = 20 \text{ V}$$

$R_x$  is one of several resistance coils made from pieces of wire drawn from an unknown metal of different lengths,  $L$ . The wires all have the same diameter of 0.6 mm. The values of  $V_{AB}$ , the voltage indicated by the multimeter (MM) and the length of wire,  $L$ , were recorded in Table 1.

**TABLE 1: RESULTS**

| Length of Coil Wire<br>$L$ (metres) | Multimeter Reading<br>$V_{AB}$ (volts) |
|-------------------------------------|----------------------------------------|
| 100                                 | 2.73                                   |
| 80                                  | 2.98                                   |
| 60                                  | 3.24                                   |
| 40                                  | 3.49                                   |
| 20                                  | 3.74                                   |
| 0                                   | 4.02                                   |

- (i) On the grid provided in Figure 3 (page 7), plot a graph of  $V_{AB}$  against  $L$  and draw your best straight line through the points. [4 marks]

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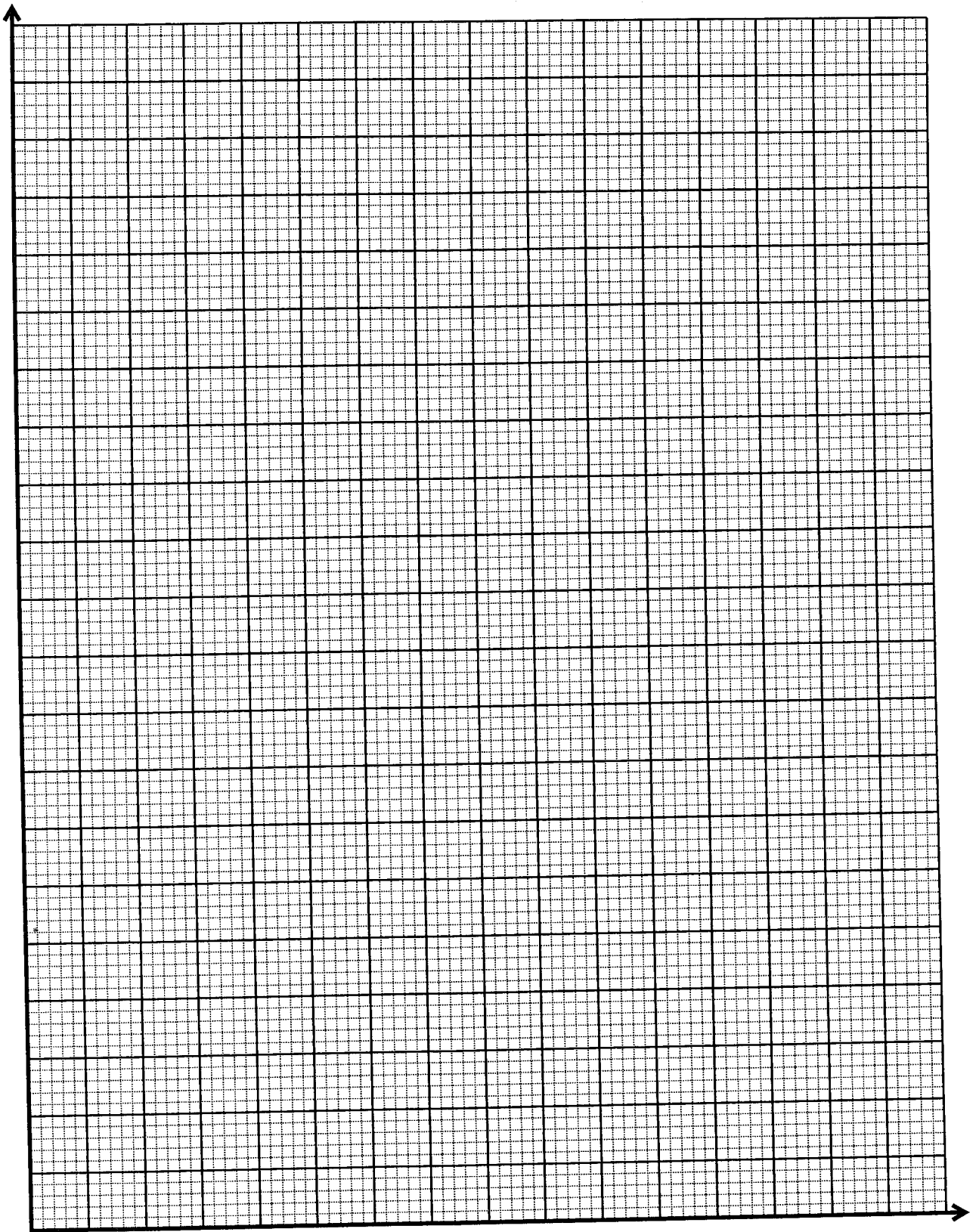


Figure 3.  $V_{AB}$  against  $L$

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2. (a) (i) Draw the circuit diagram for an inverting amplifier using an operational amplifier.
- Label EACH resistor in your circuit with the conventional term used to describe it.

[3 marks]

- (ii) State THREE properties of an ideal operational amplifier.

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[3 marks]

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- (b) A student conducted an experiment in which a variable voltage source was connected to the input of an inverting amplifier. The input voltage and the corresponding output voltage were measured using a multimeter and the readings obtained were recorded in Table 2.

**TABLE 2: MULTIMETER READINGS**

| Input Voltage<br>$V_i$ (volts) | Output Voltage<br>$V_o$ (volts) |
|--------------------------------|---------------------------------|
| -0.21                          | 2.20                            |
| 0.00                           | 0.00                            |
| 0.38                           | -3.82                           |
| 0.82                           | -8.18                           |
| 1.17                           | -12.1                           |
| 1.62                           | -15.0                           |
| 2.44                           | -15.0                           |

- (i) On the grid provided in Figure 4 (page 11), plot a graph of output voltage ( $V_o$ ) vs input voltage ( $V_i$ ).

[4 marks]

- (ii) Use your graph to calculate the gain of the amplifier.

[2 marks]

- (iii) Explain the shape of your graph.

[3 marks]

**Total 15 marks**

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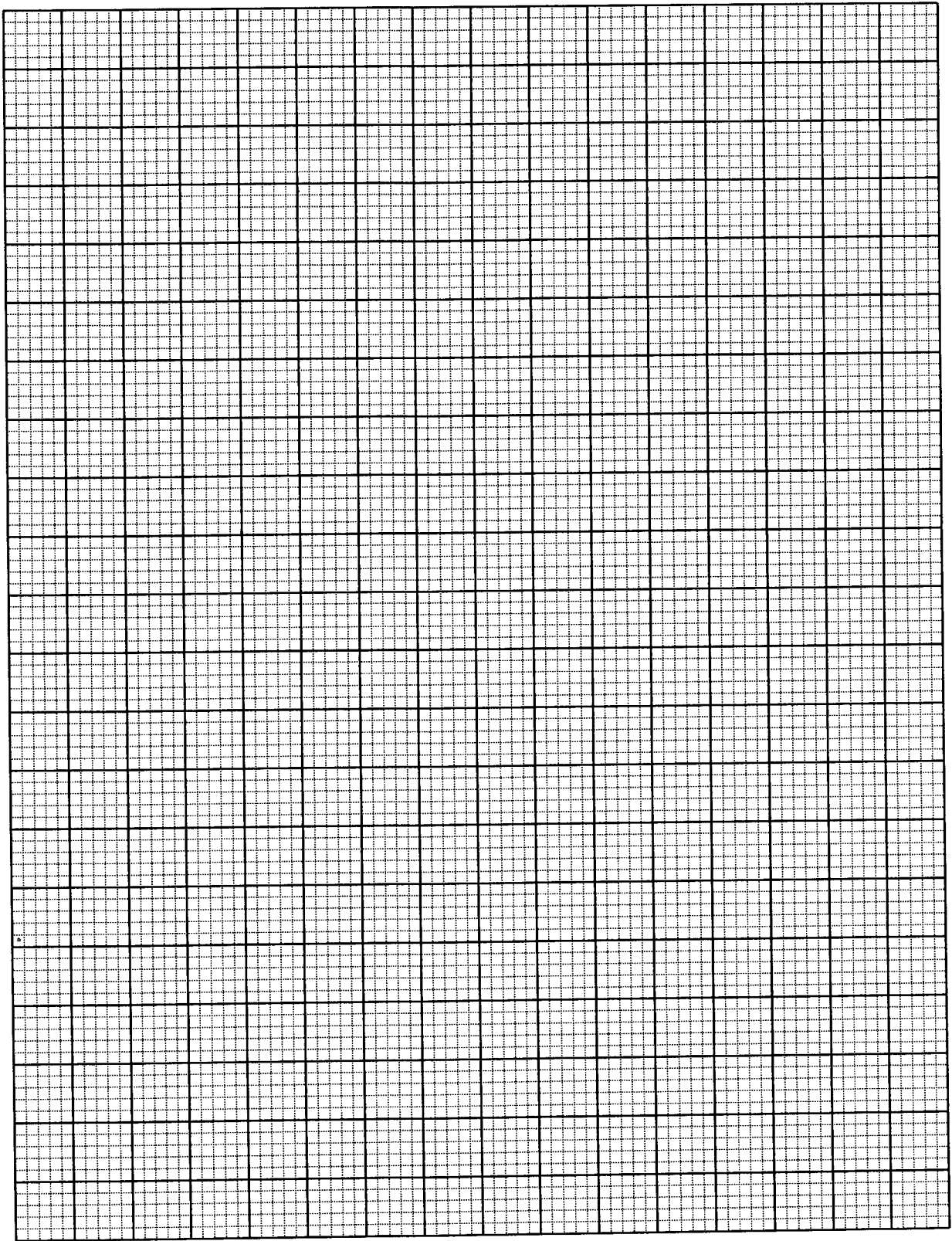


Figure 4. Output voltage vs input voltage

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3. (a) Technetium-99m is a commonly used radioisotope in nuclear medicine.

Complete Table 3 to indicate TWO properties and TWO reasons why a radioisotope such as technetium-99m can be used in this way.

**TABLE 3: RADIOISOTOPE IN NUCLEAR MEDICINE**

| Property of Radioisotope | Reason for Use in Nuclear Medicine |
|--------------------------|------------------------------------|
|                          |                                    |
|                          |                                    |

[4 marks]

- (b) (i) A sample of technetium-99m has an initial activity,  $A_0$ , of 50 GBq (Gigabecquerel). Its activity is measured at two-hour intervals and the results recorded as shown in Table 4. Complete the table by calculating and inserting appropriate values in the relevant columns.

**TABLE 4: ACTIVITY OF TECHNETIUM-99M**

| Activity<br>$A$ (GBq) | $A/A_0$ | $\ln(A/A_0)$ | Time<br>$t$ (hr) |
|-----------------------|---------|--------------|------------------|
| 39.7                  |         |              | 2.00             |
| 31.6                  |         |              | 4.00             |
| 25.1                  |         |              | 6.00             |
| 20.0                  |         |              | 8.00             |
| 15.9                  |         |              | 10.0             |
| 12.6                  |         |              | 12.0             |

[2 marks]

- (ii) On the grid provided in Figure 5 (page 13), plot a graph of  $\ln(A/A_0)$  against time,  $t$  (in hours), and draw the best straight line through the points.

[4 marks]

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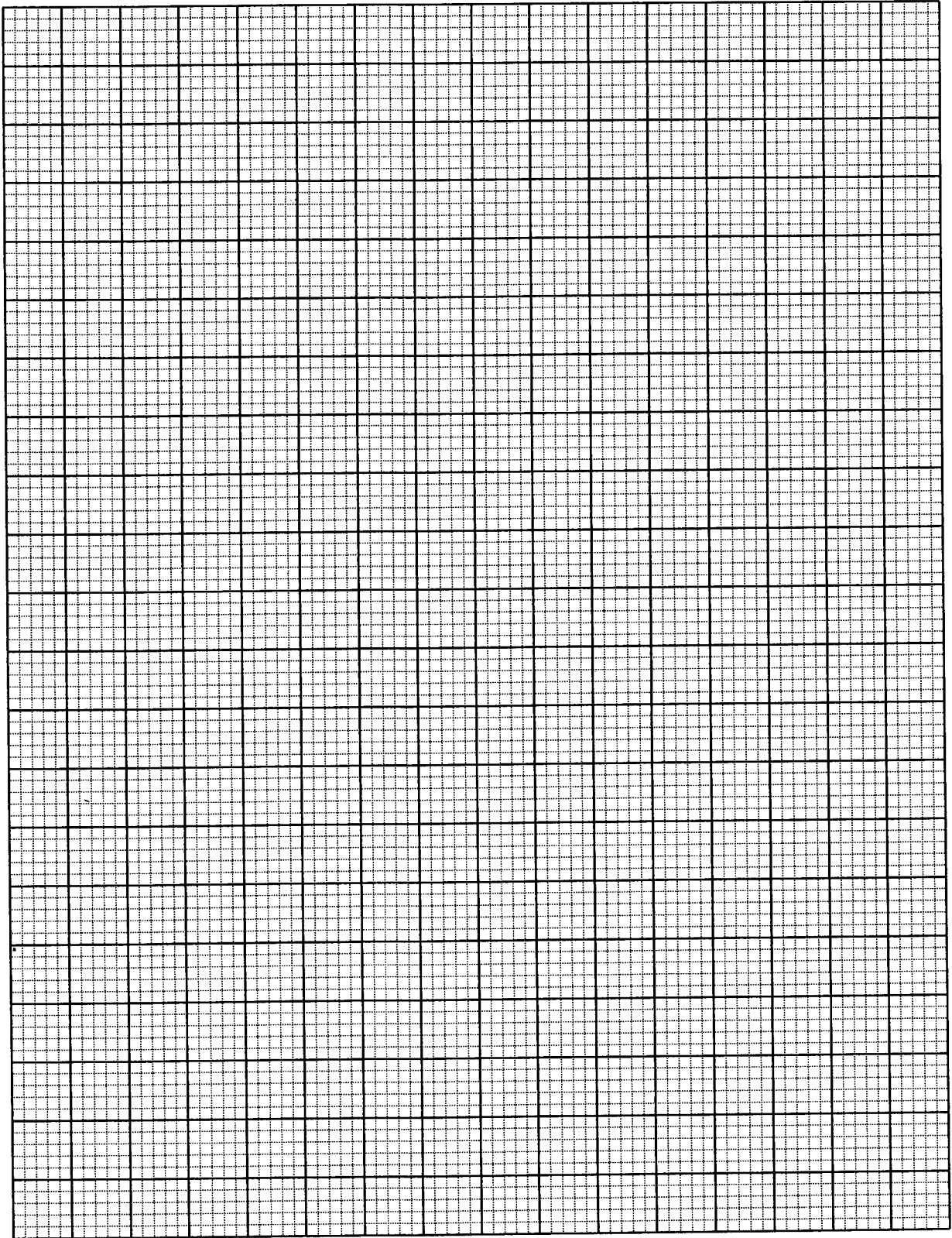


Figure 5.  $\ln (A/A_0)$  against time

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- (iii) Use your graph to find the decay constant of technetium-99m.

[3 marks]

- (iv) Use the decay constant to calculate the half-life,  $t_{1/2}$ , of technetium-99m.

[2 marks]

Total 15 marks

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SECTION B

Answer ALL questions.

Write your answers in the spaces provided.

4. (a) By using the formula  $F = BIL\sin\theta$ , define FULLY the unit of magnetic flux density, the tesla.

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[3 marks]

- (b) (i) The heaters of an electric stove dissipate 5 kW when connected to a 110 d.c. source. Calculate the current drawn from the source.

[2 marks]

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- (ii) The cable connecting the heating elements to the source consists of two parallel conductors encased in plastic insulation and separated by 0.5 cm as shown in Figure 6.

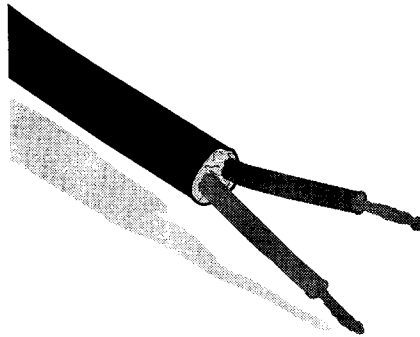


Figure 6. Two-wire high current cable

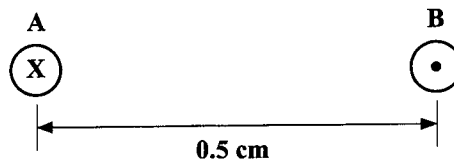


Figure 7. Current carrying conductors

Ignoring the effect of the insulation, the arrangement is approximately that of Figure 7 (not drawn to scale).

Redraw Figure 7 and add to your drawing a rough sketch of the magnetic field of the current in Conductor A. Indicate by an arrow on the diagram, the direction of the force experienced by Conductor B due to the magnetic field of the current in A.

[3 marks]

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- (c) (i) Calculate the value of the magnetic field at **B** due to the current in **A**.

[2 marks]

- (ii) Determine the magnitude and direction of the force experienced by EACH centimetre of Wire **B** due to the magnetic field of the current in Wire **A**.

[2 marks]

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- (d) The calculations carried out in (c) (i) and (c) (ii) ignore the effect of the insulating plastic sheath in which the copper conductors are encased. Deduce the effect that the plastic sheath would have on the magnetic field of the current in Wire A and give a reason for your answer.

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[3 marks]

Total 15 marks

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5. (a) Draw the truth table for a 2-input NAND gate.

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[2 marks]

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- (b) Complete Table 5 for the circuit shown in Figure 8 and hence show that this circuit is equivalent to a NAND gate.

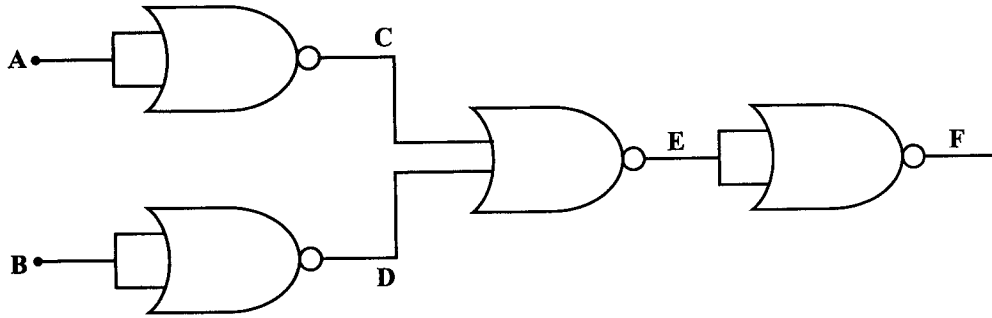


Figure 8. NAND gate equivalent

TABLE 5: LOGIC TABLE

| A | B | C | D | E | F |
|---|---|---|---|---|---|
|   |   |   |   |   |   |
|   |   |   |   |   |   |
|   |   |   |   |   |   |
|   |   |   |   |   |   |

[4 marks]

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- (c) Draw the NAND gate version of an S-R flip-flop taking care to label all inputs and outputs.

[4 marks]



- (d) The signals S and R shown in Figure 9 are simultaneously applied to the corresponding inputs of an S-R flip-flop. Determine and draw on Figure 9, the Q-output (indicates that the flip-flop is set) of the flip-flop for time intervals 1–5.

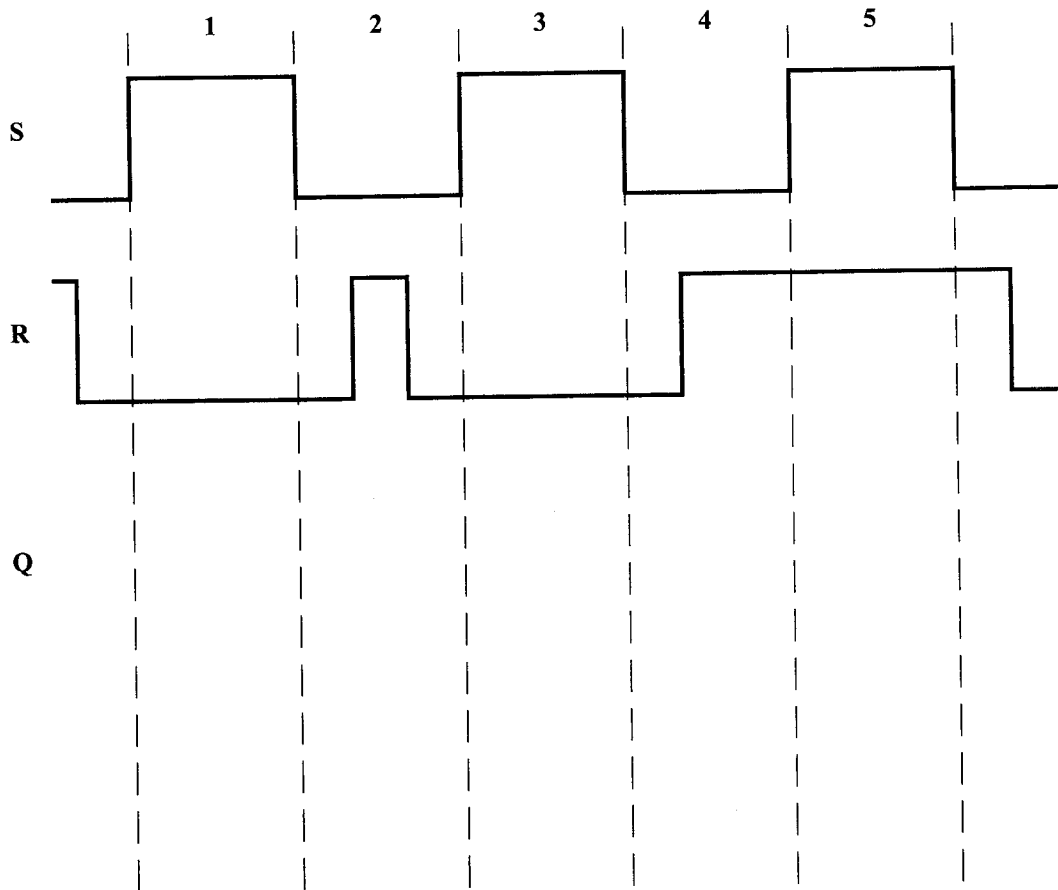


Figure 9. Timing diagram

[5 marks]

Total 15 marks

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6. (a) State TWO conclusions regarding the nature of atomic structure that were deduced from the results of the Geiger–Marsden experiment.

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[2 marks]

- (b) (i) Distinguish between the terms ‘mass defect’ and ‘binding energy’.

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[3 marks]

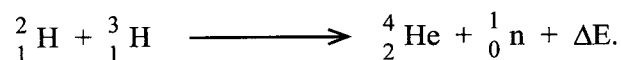
- (ii) Sketch a graph to show the relationship between binding energy per nucleon and nucleon number.

[3 marks]

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- (c) A scheme for the production of energy by nuclear fusion proposes to combine deuterium  $\left[ {}^2_1\text{H} \right]$  and tritium  $\left[ {}^3_1\text{H} \right]$  under conditions of high temperature and pressure to produce helium. The proposed reaction may be written as



The masses of the reactants and the products are:

$${}^2_1\text{H} = 2.014102 \text{ u}$$

$${}^4_2\text{He} = 4.002602 \text{ u}$$

$${}^3_1\text{T} = 3.016049 \text{ u}$$

$${}^1_0\text{n} = 1.008665 \text{ u}$$

- (i) Calculate the mass defect in kilograms.

[4 marks]

- (ii) Hence, calculate the energy in MeV which would be released in this reaction.

[3 marks]

**Total 15 marks**

**END OF TEST**

**IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.**



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# THE SCOTTSBORO BOY

